

Musubi: Octree based Lattice-Boltzmann solver for multi-physics

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Software

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Summary

Musubi is a multi-level, parallel lattice Boltzmann solver and part of the “Adaptable Poly-Engineering Simulator” (APES) suite (Klimach et al., 2023). It is working on an octree mesh that is linearized by a (Morton) space-filling curve (Sagan, 1994) and uses efficient data structures allowing for adaptive, distributed parallel simulations.

Musubi is designed to deal with huge meshes (billions of lattices) and complex geometries on large computing systems efficiently. It can be used for a wide range of application areas from electroanalysis (Masilamani, 2020) over biomedical problems (Jain, 2016) and aero-dynamic setups (Spinelli et al., 2024) to aero-acoustic simulations (Hasert, 2013).

It is written in Fortran, requiring a compiler that provides at least the Fortran 2003 standard. Parallel execution is achieved via the Message Passing Interface (MPI) of which at least the MPI-3 standard (MPI Forum, 2012) is required to support non-blocking collectives.

Statement of need

Highly resolved fluid simulations are an integral part in many scientific application areas. Due to the nonlinearity and the large amount of degrees of freedom to consider in these problems, these simulations require significant computational resources, which typically are only available in distributed parallel systems. Musubi implements the lattice Boltzmann method (LBM) with a Message Passing Interface (MPI) parallelization and a fully distributed handling of the data, avoiding bottlenecks on individual processors and enabling the scaling of the simulation to hundreds of thousands of MPI processes. Musubi is utilized to solve flow problems in the ultrasound simulator PROTEUS developed at the University of Twente (Blanken et al., 2025).

The lattice Boltzmann method

The lattice Boltzmann method (Krüger et al., 2016) employs ideas of cellular automata and can be represented at its core as a basic two step algorithm. The state of the fluid is represented by particle density functions (PDF) of a discrete velocity field. These PDFs reside on the lattices and are exchanged along the discrete velocity directions. The two steps of the algorithm are the streaming of the PDF information along velocity directions, followed by the so-called collision, computing a new PDF on each lattice. This modeling with discrete velocities also allows for a straight forward handling of complicated wall boundaries, as a simple line intersection with the wall geometry can be used to accurately model the surface. Due to these properties the method has gained popularity in the field of computational fluid dynamics over the last decades.

State of the field

There is a wide range of computational fluid dynamic methods, of which the lattice Boltzmann method represents an attractive option, due to its low number of operations and straight forward explicit implementation.

Other Open Source solvers that utilize this method are, for example, Palabos (Latt et al., 2020), OpenLB (Krause et al., 2021), waLBerla (Bauer et al., 2021) and VirtualFluids (Geier et al., 2025). They all aim at scale-resolved simulations of fluids. Musubi was developed (since 2011) in parallel or predated some of those projects due to a ground-up orientation to make use of octree meshes on massively parallel computing systems with a dedicated mesh format that allows for a distributed parallel reading from the file system. This approach allows for an automated multi-level mesh generation and avoids parallelisation bottlenecks between the mesh generation step and the simulation.

A specific domain that is addressed in Musubi and not covered by other Open Source LBM solvers, is the simulation of the Maxwell-Stefan equation for multiple species (Zudrop et al., 2017) as needed in diffusion processes that appear for example in electrodialysis applications.

Software design

Musubi implements the lattice Boltzmann method in the form of kernels that can be run on individual refinement levels of an octree mesh. It is developed within the APES-Suite (Klimach et al., 2014) of simulation tools revolving around a central Treelm library (Klimach et al., 2012) that provides the handling of this octree mesh on distributed parallel systems. This central part is implemented in a separate library, shared by the different tools in the framework. Though there still is a tight development dependency that is expressed by the use of git submodules.

The dedicated meshing tool Seeder (Harlacher et al., 2012) provides this octree mesh in a format that enables the distributed parallel reading of mesh partitions by all processes. The interpolation and transformation between the involved levels for the local refinement are separated from the kernel, allowing for an implementation of the respective methods without encumbrance by the interpolation between the different resolutions. This method was described in detail in (Hasert et al., 2014) and enables the rapid implementation of new numerical kernels. There are various collision schemes implemented (BGK, MRT, HRR, Cumulants) (Spinelli et al., 2023), which can be used on a range of stencil configurations (discrete velocity directions). It is also possible to consider the transport of particles (Vlogman, 2025) and passive scalars in the flow.

The application is designed towards deployment on a wide range of high-performance computing systems. To facilitate this, Musubi is designed with a minimal set of dependencies allowing for deployment on a variety of supercomputing systems ranging from IBM's BlueGene to NEC's SX vector systems (Qi et al., 2016). As user interface in this environment, the scripting language Lua (Ierusalimsky, 2016) is chosen, which allows for a flexible configuration of simulation setups but does not introduce complicated dependencies, as Lua is implemented in standard ANSI C and is compiled along with the project.

Research impact statement

Musubi has been successfully deployed in the simulation of scale-resolved fluid problems for a variety of problems. Originally, a main funding source for the software was the German HISEEM project that aimed at the investigation of effective electrodialysis processes for seawater desalination (Johannink et al., 2015). Another main focus in the development is put on biomedical application (Jain et al., 2016), which was originally supported by the European THROMBUS project (Zimny et al., 2013). In the biomedical domain Musubi now is also used

85 as the fluid simulation tool in the ultrasound simulator PROTEUS developed at the University
86 of Twente (Blanken et al., 2025).

87 While the two aforementioned application domains primarily are concerned with the simulation
88 of liquids, there are also applications to gaseous fluids, for example in the simulation of human
89 upper airway aerodynamics (Hebbink et al., 2022). Musubi's application extends beyond these
90 domains and has been used in general aerodynamic simulations (Spinelli et al., 2024), as well
91 as in the domain of aero-acoustics (Qi et al., 2015), where the resolution of multiple spatial
92 scales plays an important role. The distributed handling of octree meshes is well suited in
93 these settings as it allows for the resolution of the scales that need to be resolved.

94 AI usage disclosure

95 No generative AI tools were used in the development of this software, the writing of this
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